

Classification Review

1. Precision

Precision measures how many of the predicted samples for a class were actually correct.

Formula: $\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$

Where TP = True Positives and FP = False Positives.

For the Action class: TP = 1, FP = 0

$\text{Precision} = 1 / (1 + 0) = 1.00$

The same applies to Comedy, Crime, and Romance.

2. Recall

Recall measures how many actual samples of a class were correctly identified.

Formula: $\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$

Where TP = True Positives and FN = False Negatives.

For Action: TP = 1, FN = 0

$\text{Recall} = 1 / (1 + 0) = 1.00$

3. F1-Score

F1-Score is the harmonic mean of Precision and Recall.

Formula: $\text{F1} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$

For Action:

$\text{F1} = 2 \times (1 \times 1) / (1 + 1)$

$\text{F1} = 2 / 2 = 1.00$

The same result is obtained for all classes.

4. Accuracy

Accuracy measures the overall correctness of the model.

Formula: $\text{Accuracy} = \text{Correct Predictions} / \text{Total Predictions}$

From the report:

Correct Predictions = 4

Total Predictions = 4

$\text{Accuracy} = 4 / 4 = 1.00$ (100%)

Final Review

The model correctly classified all four test samples (Action, Comedy, Crime, and Romance). Since there were no false positives or false negatives, Precision, Recall, and F1-Score for every class are

1.00. The overall accuracy is 100%, indicating perfect performance on the given test dataset. However, because the support value is only 4 samples, evaluation on a larger dataset is recommended to confirm the model's real-world performance and generalization ability.